

HARMILAP SINGH DHALIWAL

AI Evaluation & Quality · Model Risk & Responsible AI · Agentic AI in Regulated Finance

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AI & Data Science Lead in RBC's CFO Group, where measured quality is the control that lets AI ship safely and fast in a regulated bank. Across eight-plus years, including leading a technical team, evaluation has been the gate on the systems I ship: from a multi-indicator framework that qualified 84 models before a live power-plant control loop, to the LLM-as-judge plus structured human review that carried RBC's first bank-wide production AI agent through Model Risk Management. Three-plus of those years are production AI in regulated finance, the most recent year-plus on agentic LLM systems evaluated across their outputs, tool-call trajectories, and runtime behavior. The throughline is translating model risk, responsible AI, and business expectations into measurable criteria, repeatable methods, and audit-ready evidence.

PROFESSIONAL EXPERIENCE

AI & Data Science Lead, Finance — Royal Bank of Canada

April 2025 – Present

CFO Group, Enterprise Finance · Toronto, ON

Hire, develop, and coach a technical team: 1 Senior AI Scientist (direct) plus 2 interns onboarded May 2026 (9 cumulative, peak 5). Contribute to AI/ML hiring, screening, and performance reviews across CFO Group since 2023. Define evaluation standards, acceptance criteria, and reviewer guidance; lead cross-functional production deployment with Global Functions Technology via CI/CD on OpenShift. Hands-on across ~70% of project work.

Evaluation & Production AI

- › Designed the evaluation behind PAR Assist, RBC's first bank-wide production AI agent (pilot April, bank-wide May 2026): LLM-as-judge plus structured human review producing the quality and safety evidence for Model Risk Management sign-off.
- › Evaluate agent behavior across outputs, tool-call trajectory, and runtime: a LangGraph state graph and a typed MCP tool layer as the logged action boundary, with custom tracing and structured logging.
- › Built the versioned regression eval set that gates Astraeus foundation-model upgrades (multi-stage LLM analytics, production November 2025), proving a clean Command-A to GPT-4.1 migration with no quality loss.
- › Engineered defense-in-depth runtime gating (relevance, injection, scope, privilege) between Astraeus stages as failure-mode coverage; it serves headcount and cost queries across ~40K cost centres and ~9K rollups.
- › Built the Aegis v2 benchmarking module solo in two weeks while running Astraeus and the 2025 intern program in parallel; productionized by my direct report.

Evaluation Standards, Model Risk & Leadership

- › Carry production AI through Model Risk Management and responsible-AI review end-to-end, owning the evidence: measurable acceptance criteria, LLM-as-judge, human review, and post-deployment monitoring, all built in-house with no third-party eval libraries.
- › Extending PAR Assist's single-agent drafter into a config-driven multi-agent drafting framework, where a new product is a configuration rather than a rebuild; first target is a regulated financial-governance memo.
- › Built and shipped ARGUS, an analytics-agent prototype for senior bank leadership, solo over one weekend within RBC's fixed internal AI platform; demoed to multiple SVPs.
- › Lead the Amplify intern program and the team's engineering practices: PR-first review, agentic-design and RAG architecture reviews, and reviewer guidance that lets juniors ship production code.

Enterprise Finance Senior Data Scientist — Royal Bank of Canada

September 2022 – April 2025

CFO Group, Enterprise Finance · Toronto, ON

- › Built Aegis v1 end-to-end solo: its curation, validation, and matching pipeline over quarterly-shifting Big-6 financial-package schemas cleared the data-quality bottleneck blocking timely peer benchmarking. CFO One RBC Team Award (2025).
- › Cut commodity-tax return prep from months to 90 minutes across ~\$250M GST and ~\$350M PVAT with a PySpark-on-CDP pipeline over GL Journal data. CFO Group RBC Quarterly Team Award (Q4 2023).

- › Built EDS Automation, a PAR actual-vs-planned financial comparison system (custom Python with Dataiku as scheduler, Tableau dashboards), widely adopted across the finance team.

Machine Learning Engineer — Quantiphi Inc.

October 2021 – September 2022

Toronto, ON

- › Lifted document-understanding accuracy from ~70% to 99.95% on the Humana pipeline (GCP Document AI OCR, OpenCV, Random Forest), with per-field accuracy benchmarked against human-labeled ground truth before each release.

Data Scientist — Tata Consultancy Services

August 2016 – November 2019

Pune, India

- › Designed the multi-indicator evaluation framework (k-fold, R^2 , RMSE, MAPE, fold variance) that qualified all 84 regression models on accuracy and robustness, the gate that kept brittle models out of the optimization layer.
- › Architected the Combustion Tuning Digital Twin end-to-end (90+ sensor ETL, 84 regression models, Particle Swarm Optimization closing the control loop): \$3M annual savings and measurable NOx/SOx/CO reduction at a 900MW coal plant.
- › Won 2nd of 600 in TCS's CV Hackathon for a deep-learning math-notation detector (Innovation Pride Award, 2019).

TECHNICAL SKILLS

AI Evaluation & Measurement	LLM-as-judge design, rule-based / deterministic checks, human evaluation protocols and reviewer guidance, output / trajectory / runtime evaluation, hallucination / factuality / safety assessment, scorer calibration and robustness, eval datasets and scorecards (curation, validation, versioning, lineage, reuse), regression / golden suites on model upgrades, model-quality benchmarking
Model Risk & Responsible AI	Model Risk Management (MRM) review and evidence packages, responsible-AI review, governance-to-measurable-control translation, measurable acceptance criteria, audit-ready evidence, audit trails and least-privilege enforcement across multi-stage pipelines, regulated-finance quality bar
Agentic AI & LLMops	Agentic state graphs (LangGraph), typed MCP tool layer as logged action boundary, config-driven multi-agent drafting framework, multi-stage LLM pipelines, defense-in-depth runtime gating, LLM-as-router then dense RAG, foundation-model-agnostic (GPT-4.1, Command-A), custom tracing and structured eval logging, post-production monitoring, CI/CD on OpenShift
Data Science Foundations	Python, SQL, statistics, experimentation, data curation, regression, k-fold validation, PySpark on Cloudera Data Platform, ETL, scikit-learn, Particle Swarm Optimization

EDUCATION & CERTIFICATIONS

Post Graduate Professional Certificate, Big Data Analytics

January 2021 – August 2021

Georgian College, Barrie, Ontario

Bachelor of Engineering, Electronics & Communications Engineering

July 2012 – June 2016

Thapar University, Patiala, India

CERTIFICATIONS DeepLearning.AI Deep Learning Specialization · TensorFlow Developer Specialization · Johns Hopkins Data Science Specialization

AWARDS & RECOGNITION

CFO One RBC Team Award (2025, LLM/AI & Aegis benchmarking) · **CFO Group RBC Quarterly Team Award** (Q4 2023, Commodity Tax automation) · **Innovation Pride Award, TCS** (2019, 2nd of 600, CV Hackathon).